

Retake Preparation: Part 3.5

Quiz 2 Uncovered & Alternate Topics

Section 1: The Relations Between Distributions

Your professor explicitly noted: “*You should be aware of the relations between these distributions.*” These are high-yield, short-answer questions designed to test how standard distributions are structurally connected.

The 5 Classic Relational Questions:

1. **Question:** What is the relationship between the **Standard Normal** (Z) and the **Chi-Square** (χ_1^2) distribution?

Answer: If $Z \sim N(0, 1)$, then its square follows a Chi-Square distribution with 1 degree of freedom:

$$Z^2 \sim \chi_1^2$$

2. **Question:** How is the **Student’s t -distribution** constructed?

Answer: It is constructed by taking a Standard Normal variable $Z \sim N(0, 1)$ and dividing it by the square root of an independent Chi-Square variable $U \sim \chi_n^2$ divided by its degrees of freedom:

$$T = \frac{Z}{\sqrt{U/n}} \sim t_n$$

3. **Question:** How is the **F -distribution** constructed?

Answer: It is constructed by taking the ratio of two independent Chi-Square variables ($U \sim \chi_m^2$ and $V \sim \chi_n^2$), each divided by their respective degrees of freedom:

$$W = \frac{U/m}{V/n} \sim F_{m,n}$$

4. **Question:** What is the relation between the **Exponential** and **Poisson** distributions?

Answer: If events occur over time as a Poisson process with rate α , the number of events in an hour follows a Poisson distribution with $\lambda = \alpha$. However, the **waiting time between those consecutive events** follows an Exponential distribution with parameter $\lambda = \alpha$.

5. **Question:** What is the relation between the **Gamma** and **Exponential** distributions?

Answer: The Exponential distribution is a special case of the Gamma distribution where the shape parameter $\alpha = 1$. Furthermore, the sum of r independent, identical Exponential (λ) variables follows a Gamma distribution with parameters $\alpha = r$ and λ .

Section 2: Finding a CDF from a PDF (Fundamental Theorem of Calculus)

Question (HW 9, Q4 style): A continuous random variable X has a probability density function:

$$f(x) = \begin{cases} \frac{3}{8}x^2 & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the Cumulative Distribution Function $F(x) = P(X \leq x)$ for all x , and calculate $P(X \leq 1)$ using $F(x)$.

Solution:

1. **Formula Applied (Fundamental Theorem of Calculus):** To find the cumulative function $F(x)$, we integrate the density function from the lowest boundary (0) up to a variable point x :

$$F(x) = \int_0^x f(t) dt$$

2. **Why this is used:** The CDF represents the accumulated area under the PDF curve from $-\infty$ up to x . By the FTC, the derivative of $F(x)$ is the PDF $f(x)$.

3. **Calculation of $F(x)$:** For $x < 0$, $F(x) = 0$.

For $0 \leq x \leq 2$:

$$F(x) = \int_0^x \frac{3}{8}t^2 dt = \frac{3}{8} \left[\frac{t^3}{3} \right]_0^x = \frac{3}{8} \left(\frac{x^3}{3} - 0 \right) = \frac{x^3}{8}$$

For $x > 2$, $F(x) = 1.0$.

4. **Using $F(x)$ to find $P(X \leq 1)$:**
Since 1 is inside the interval $[0, 2]$:

$$P(X \leq 1) = F(1) = \frac{1^3}{8} = \frac{1}{8} = 0.125$$

Section 3: Gamma Distribution Simple Integration (HW Distributions, Q13)

Question: Let X have a Gamma density with parameters $\alpha = 2, \lambda = 1$:

$$f(x) = xe^{-x}, \quad x > 0$$

Find the probability $P(X > 1)$ using integration by parts.

Solution:

1. **Formula Applied (Integration by Parts):** To find $P(X > 1)$, we integrate the density from 1 to ∞ :

$$P(X > 1) = \int_1^{\infty} x e^{-x} dx$$

We use: $\int u dv = uv - \int v du$. Let $u = x \implies du = dx$ and $dv = e^{-x} dx \implies v = -e^{-x}$.

2. **Calculation:**

$$\int x e^{-x} dx = -x e^{-x} - \int -e^{-x} dx = -x e^{-x} - e^{-x} = -e^{-x}(x + 1)$$

Now we evaluate this from 1 to ∞ :

$$P(X > 1) = \left[-e^{-x}(x + 1) \right]_1^{\infty}$$

For the upper limit (∞): $\lim_{x \rightarrow \infty} -e^{-x}(x + 1) = 0$ (exponential decay dominates).

For the lower limit (1): $-e^{-1}(1 + 1) = -2e^{-1}$.

$$P(X > 1) = 0 - (-2e^{-1}) = \frac{2}{e} \approx \mathbf{0.7358}$$

Section 4: Memoryless Property of Exponential (HW 9, Q3)

Question: Suppose that the lifetime of a certain machine battery has an exponential distribution with parameter $\lambda = 0.2$. Find the conditional probability $P(X > 8 \mid X > 3)$.

Solution:

1. **Formula Applied (The Memoryless Property):** For any exponentially distributed continuous variable X [1]:

$$P(X > s + t \mid X > s) = P(X > t)$$

2. **Why this is used:** The exponential distribution is the only continuous distribution with no memory. If the battery has already survived 3 years ($s = 3$), the probability of it surviving for another 5 years ($t = 5$, making total lifetime > 8 years) is identical to its basic probability of surviving for 5 years [1].

3. **Calculation:**

$$P(X > 8 \mid X > 3) = P(X > 5)$$

Using the exponential survival formula ($P(X > x) = e^{-\lambda x}$):

$$P(X > 5) = e^{-0.2 \times 5} = e^{-1} \approx \mathbf{0.3679}$$